

SEG3125 Assignment 3: Memory Game — Games for Exploring Human Cognition

Link to github: <https://github.com/tonylilui/seg3125-assignment1-portfolioentrypage-tonyli-300363068.git>

High-Fidelity Prototype Report

Designer: Tony Li
Student Number: 300363068
Date: June 12, 2026
Course: SEG3125 - User Interface Analysis and Design
Submission: Assignment 3 — Memory Game

DESIGNER INFORMATION

- **Name:** Tony Li
- **Student Number:** 300363068
- **Institution:** University of Ottawa
- **Course:** SEG3125 - Analysis and Design of User Interfaces

GAME CONCEPT

Memory Card Matching Game (Un jeu de mémoire)

Game Type: Pair-matching memory game — a grid of face-down cards where the player flips two at a time to find matching pairs.

Game Description: The Memory Game challenges users to test and improve their short-term visual memory by finding matching pairs of icons hidden behind a grid of cards. The game features:

- **Three difficulty levels** (Easy 4x4, Medium 4x5, Hard 6x6) that scale the number of pairs
- **Card flip animations** with 3D perspective transforms for tactile feedback
- **Move counter and timer** to track performance and encourage improvement
- **Efficiency scoring system** (S/A/B/C/D grades) based on move efficiency relative to optimal play
- **Progressive disclosure** — instructions appear on first visit, then disappear

Why This Game: A memory card matching game was selected because it:

1. Directly exercises short-term visual-spatial memory — the core cognitive skill the assignment targets
2. Has universal appeal and is instantly understandable across cultures and age groups
3. Allows clear application of Gestalt principles (proximity, similarity, closure, continuity)
4. Provides natural opportunities for iterative design improvements
5. Builds on Assignment 1 (portfolio site) and Assignment 2 (service site) by continuing the React/TypeScript/Bootstrap stack

3. USER-CENTERED DESIGN (UCD) APPROACH

Target Users

Based on the portfolio context, the primary users are:

User 1: Students & Learners

- **Demographics:** Ages 18–25, university students
- **Tech Comfort:** High — native digital generation
- **Memory Profile:** Moderate — familiar with memory games but not experts
- **Goal:** Practice memory skills in a fun, visually appealing way

User 2: Professionals Seeking Brain Training

- **Demographics:** Ages 25–45, working professionals
- **Tech Comfort:** High — use digital tools daily
- **Memory Profile:** Variable — may be concerned about cognitive decline from screen time
- **Goal:** Quick mental exercise during breaks

Design Process (Iterative UCD Cycle)

1. **Research:** Analyzed existing memory games (HelpfulGames, MathIsFun Simon Says, MemoryLossTest) to identify common patterns and UI shortcomings
2. **Ideation:** Brainstormed card-based matching as the core mechanic — the most intuitive and widely understood memory game format

3. **Prototyping:** Built interactive React prototype with:
 - Card flip animations (closure principle)
 - Grid layout (continuity principle)
 - Color-coded scoring (similarity principle)
4. **Evaluation:** Applied Nielsen's heuristics and Gestalt principles to validate design decisions

Personas Applied to Game Design

Persona Need	Design Response
Quick session play	Three difficulty levels (2–10 min per game)
Clear feedback	Animated match confirmation + score grade
Accessibility	Large touch targets, high contrast, responsive layout
Motivation	Efficiency scoring encourages replayability

GESTALT PRINCIPLES APPLIED

Proximity (Elements near each other are perceived as related)

Application:

- **Stats bar:** Moves, time, and matches are grouped in a single horizontal bar with consistent spacing, making them read as one unit
- **Difficulty buttons:** Grouped together with uniform styling to indicate they control the same function
- **Win modal stats:** Three columns (moves, time, efficiency) are evenly spaced within a shared background container

Code Reference: `MemoryGame.css` — `.stats-bar`, `.difficulty-selector`, `.win-stats` use flexbox with consistent `gap` values

Similarity (Similar elements are perceived as related)

Application:

- **All cards:** Identical size, border-radius, and shadow regardless of state (flipped/matched/unflipped)
- **Interactive elements:** Buttons share consistent hover effects (`translateY` + shadow)
- **Score grades:** Color-coded consistently — green (S/A), blue (B), yellow (C), red (D)
- **Matched cards:** All matched cards receive the same green border glow

Code Reference: `MemoryGame.css` — `.card`, `.btn`, `.score-grade` with color mapping

Closure (Mind fills in missing information)

Application:

- **Card backs:** The patterned card back creates a "hidden" state — the mind anticipates what's underneath
- **3D flip animation:** The card appears to physically rotate, completing the mental model of a physical card deck
- **Icon recognition:** Even partially visible icons are recognized by their distinctive shapes (brain, star, moon)

Code Reference: `MemoryGame.css` — `.card-inner` with `transform-style: preserve-3d` and `backface-visibility: hidden`

Continuity (Eye follows a continuous path)

Application:

- **Grid layout:** Cards are arranged in a clean grid that guides the eye left-to-right, top-to-bottom — matching natural reading patterns
- **Progressive reveal:** As cards flip, the eye naturally tracks which icons have been seen and where they might be
- **Gradient background:** The purple-blue gradient flows diagonally, creating visual depth without distracting from the game board

Code Reference: `MemoryGame.css` — `.game-board` with CSS Grid, `.memory-game` with diagonal gradient

Figure-Ground (Separation of foreground and background)

Application:

- **Cards on board:** White card fronts stand out clearly against the purple gradient background
- **Win modal overlay:** Semi-transparent dark backdrop with blur separates the modal from the game
- **Instructions panel:** White panel on gradient background creates clear content hierarchy
- **Stats bar:** Frosted glass effect (`backdrop-filter: blur`) keeps stats readable while maintaining visual connection to background

Code Reference: `MemoryGame.css` — `.win-modal-overlay` with `backdrop-filter`, `.stats-bar` with frosted glass

Symmetry (Balanced composition)

Application:

- **Centered layout:** All major elements (title, controls, board, modal) are horizontally centered
- **Grid symmetry:** Square cards create a balanced grid regardless of difficulty level
- **Win modal stats:** Three columns with equal spacing and aligned typography

Code Reference: `MemoryGame.css` — `max-width` containers with `margin: 0 auto`, `flexbox justify-content: center`

ATTENTION & PERCEPTION PRINCIPLES

Selective Attention

- **Minimal distractions:** No ads, pop-ups, or unrelated content during gameplay
- **Visual hierarchy:** The game board is the largest element; secondary controls are smaller and less prominent
- **Focus state:** Card hover effects (lift + shadow) draw attention to interactive elements without overwhelming

Working Memory Limits (Miller's Law: 7±2 items)

- **Icon selection:** Maximum 18 pairs (Hard mode) — within working memory capacity for pattern recognition
- **Progressive difficulty:** Easy (8 pairs) → Medium (10 pairs) → Hard (18 pairs) scaffolds cognitive load

Hick's Law (More choices = longer decision time)

- **Three clear options:** Only three difficulty levels — enough variety without overwhelming
- **One primary action per screen:** Flip a card, view results, or choose difficulty — never all at once

Fitts's Law (Larger/closer targets are easier to interact with)

- **Card size:** Cards are generously sized (minimum 60px on mobile) for easy tapping
- **Button placement:** Primary actions (Play Again, Reset) are large and centrally located

DESIGN PATTERNS APPLIED

Organization Patterns

- **Progressive disclosure:** Instructions shown only on first visit; difficulty selector hidden until needed
- **Information hierarchy:** Title → Controls → Game Board → Results (top-to-bottom flow)

Input Patterns

- **Direct manipulation:** Click/tap cards directly — no intermediate dialogs
- **Toggle:** Card flip is a toggle between face-down and face-up states
- **Segmented control:** Difficulty buttons act as a segmented control for game mode

Navigation Patterns

- **Breadcrumb:** "← Back to Portfolio" provides clear return path
- **Reset:** Inline reset button avoids navigation away from the game
- **Modal actions:** Win screen offers Play Again and Change Difficulty — both keep user in context

TECHNICAL IMPLEMENTATION

Stack

- **Framework:** React 18 with TypeScript
- **Styling:** Custom CSS with CSS Grid, Flexbox, CSS animations
- **Build:** Create React App (react-scripts 5.0.1)
- **Dependencies:** Bootstrap 5 (for portfolio integration)

Key Technical Decisions

Decision	Rationale

CSS Grid for game board Decision	Responsive layout that adapts to difficulty levels Rationale
CSS 3D transforms for card flip	Smooth, hardware-accelerated animation without libraries
React state management (useState)	Simple, predictable state flow for game logic
useEffect for match detection	Automatic evaluation after state changes
useCallback for handlers	Prevents unnecessary re-renders of click handlers

Game Logic Flow

1. Player selects difficulty → Board generates with shuffled cards
2. Player clicks card → Card flips (animation)
3. Player clicks second card → Second card flips
4. System checks match:

- Match → Both cards glow green, remain face-up

- No match → Both cards flip back after 1 second
5. All pairs found → Win modal with score and stats

COMPARISON TO EXISTING MEMORY GAMES

Feature	HelpfulGames	MathIsFun Simon	MemoryLossTest	My Game
Card matching	✓	✗ (Simon)	✓	✓
Multiple difficulties	✓	✓	✗	✓
Modern UI	✗ (dated)	✗ (dated)	✗ (dated)	✓
Score grading	Basic	✗	✗	S/A/B/C/D
Responsive design	Partial	Partial	Partial	✓
Smooth animations	✗	Basic	✗	✓ (3D flip)
Clear feedback	Moderate	Moderate	Moderate	✓ (animated)

Key Differentiator: The modern UI, smooth 3D card animations, and efficiency-based scoring provide a significantly better user experience than the dated interfaces of existing memory game sites.

LEARNING OUTCOMES

What I Learned

1. **Gestalt principles in practice:** Applying proximity, similarity, closure, continuity, figure-ground, and symmetry to a real project deepened my understanding beyond theory
2. **Cognitive load management:** Designing difficulty levels that respect working memory limits (Miller's Law) was challenging but rewarding
3. **Animation for feedback:** 3D card flip animations provide crucial tactile feedback that improves the user's mental model of the game
4. **Score design:** Creating a meaningful scoring system (efficiency-based S/A/B/C/D grades) adds replayability without complexity

Challenges Faced

1. **Balancing difficulty:** Ensuring Hard mode (18 pairs) is challenging but not frustrating required careful tuning
2. **Animation timing:** The 500ms match delay and 1000ms no-match delay needed testing to feel responsive without being jarring
3. **Responsive grid:** Making the card grid work across all screen sizes while maintaining square aspect ratios

FUTURE IMPROVEMENTS

- **Sound effects:** Add subtle flip and match sounds for enhanced feedback
 - **High scores:** Local storage persistence for replay comparison
 - **Hint system:** Reveal all cards briefly as a one-time hint
 - **Timed mode:** Challenge players to complete within a time limit
 - **Accessibility:** Keyboard navigation, screen reader support, reduced motion option
-

CONCLUSION

This memory game assignment successfully applied User-Centered Design principles and Gestalt laws of perception to create an engaging, visually polished memory card matching game. The design addresses the needs of both casual players and brain-training enthusiasts through progressive difficulty, clear feedback, and modern UI aesthetics that surpass existing solutions in the space.

The project builds on Assignment 1 (portfolio site) and Assignment 2 (service site) by continuing to demonstrate proficiency in React/TypeScript development, Bootstrap integration, and user-centered design thinking.

Files:

- `src/components/MemoryGame.tsx` — Game component
- `src/components/MemoryGame.css` — Styles with Gestalt principles
- `case-studies/memory-game.html` — Updated to link to the live game